

Demonstration of ECR and ECS

Step-1: Login to your AWS IAM user account

Step-2: Go to ECR->click on create repository

Create private repository

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
559050212593.dkr.ecr.ap-south-1.amazonaws.com/
16 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./

Image tag settings [info](#)

Image tag mutability
Choose the tag mutability setting.
☐ Mutable
Image tags can be overwritten.
☒ Immutable
Image tags can't be overwritten.

Immutable tag exclusions
Tags that match these filters will be mutable (can be overwritten). Using wildcards (*) will match zero or more image tag characters.

Filters must only contain letters, numbers, and special characters [_*]. Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list. [Add Filter](#)

Encryption settings [info](#)

Encryption configuration
By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.
☒ AES-256
Industry standard Advanced Encryption Standard (AES) encryption
☐ AWS KMS
AWS Key Management Service (KMS)

Image scanning settings - deprecated

Deprecation warning
This setting has been moved to a registry level configuration with repository filtering. [Learn more](#)

Scan on push
Enable scan on push to have each image automatically scanned after being pushed to a repository. If disabled, each image scan must be manually started to get scan results.
☒

[Cancel](#) [Create](#)

Step-3: Open command prompt and check if cli is installed or not using command `aws --version`

```
bsowm@LAPTOP-78EC904Q: /mnt/c/Windows/System32
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>wsl
bsowm@LAPTOP-78EC904Q:/mnt/c/Windows/System32$ aws --version
aws-cli/2.34.32 Python/3.14.4 Linux/6.6.87.2-microsoft-standard-WSL2 exe/x86_64.ubuntu.24
bsowm@LAPTOP-78EC904Q:/mnt/c/Windows/System32$
```

Step-4: Now connect to aws using command `aws configure`. Then after install docker inside wsl following below commands

`sudo apt update`

`sudo apt install docker.io -y`

`docker --version`

Step-5: start running the docker using below command

```
sudo service docker start
```

```
sudo docker ps
```

Step-6: Instead of using sudo everytime lets fix the permission using below command

```
sudo usermod -aG docker $USER
```

Then after try running `docker ps`. If its works correctly permissions are fixed.

Step-6: create a directory and then create a html file as below

```
mkdir ~/ecr-demo
```

```
cd ~/ecr-demo
```

```
nano index.html
```

```
cat index.html
```

```
bsowm@LAPTOP-78EC904Q: ~/ecr-demo
GNU nano 7.2 index.html *
<!DOCTYPE html>
<html>
<body>
<h1>Hello from ECR!</h1>
</body>
</html>
```

Step-7: Now create a docker file, build image, test image locally.

```
nano Dockerfile
```

```
FROM nginx:latest
```

```
COPY index.html /usr/share/nginx/html/index.html
```

```
EXPOSE 80
```

Save and exit Dockerfile after entering the above contents.

```
docker build -t sowmya-demo-repo .
```

```
docker images
```

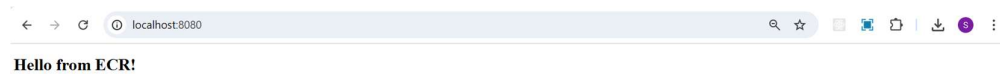
```
bsowm@LAPTOP-78EC904Q:~/ecr-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	5aca99593157	241MB	66MB	
sowmya-demo-repo:latest	dad44226ad7d	238MB	63.1MB	

```
docker run -d -p 8080:80 sowmya-demo-repo
```

```
docker ps
```

open browser and paste <http://localhost:8080>



Step-8: Login to ECR and push the image to ECR using below commands.

```
aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 559050212593.dkr.ecr.ap-south-1.amazonaws.com
```

Here I have given my credentials to login.

```
Then Tag the image using docker tag sowmya-demo-repo:latest 559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo:latest
```

Verify using `docker images` you will see the images created.

Push the image to ECR using command

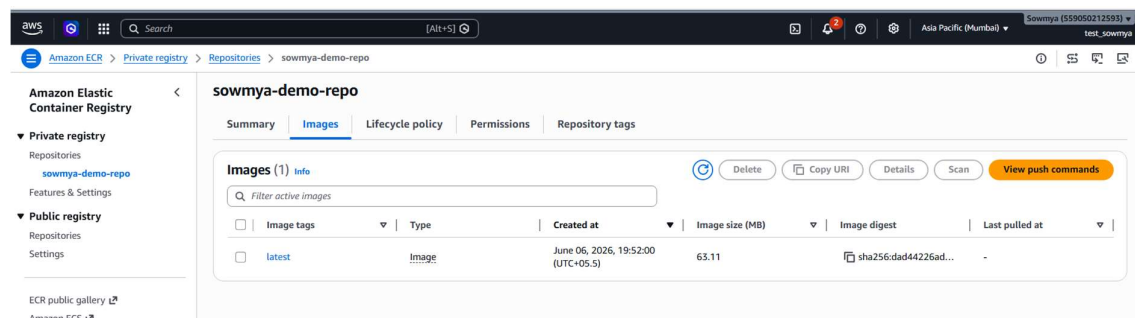
```
docker push 559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo:latest
```

```
Login Succeeded
bsowm@LAPTOP-78EC904Q:~/ecr-demo$ docker tag sowmya-demo-repo:latest 559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo:latest
bsowm@LAPTOP-78EC904Q:~/ecr-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo:latest	dad44226ad7d	238MB	63.1MB	U
nginx:latest	5aca99593157	241MB	66MB	
sowmya-demo-repo:latest	dad44226ad7d	238MB	63.1MB	U

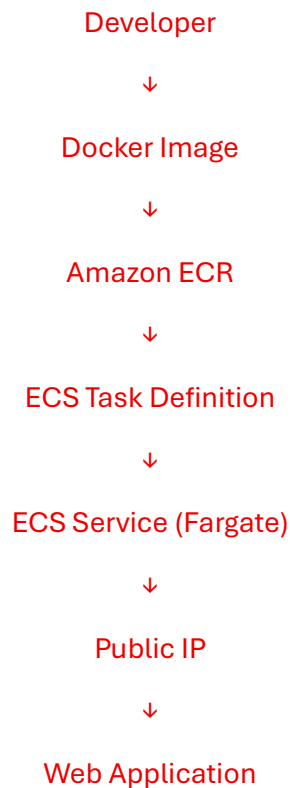
```
bsowm@LAPTOP-78EC904Q:~/ecr-demo$ docker push 559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo:latest
The push refers to repository [559050212593.dkr.ecr.ap-south-1.amazonaws.com/sowmya-demo-repo]
13fd728be9eb: Pushed
45381ecb0e2f: Pushed
5b4d6ff92fc4: Pushed
830625e1ac85: Pushed
5431d0092ffd: Pushed
1602a2a52f8a: Pushed
7f8b1a2b17d8: Pushed
b4a248c845e5: Pushed
latest: digest: sha256:dad44226ad7d45f46439e6a6a3da28b3e5033e94fe61489fb0b6e2198c9d234b size: 2043
```

Step-9: Go to AWS console->ECR->open the repository created you will able to see the image pushed.



ECS

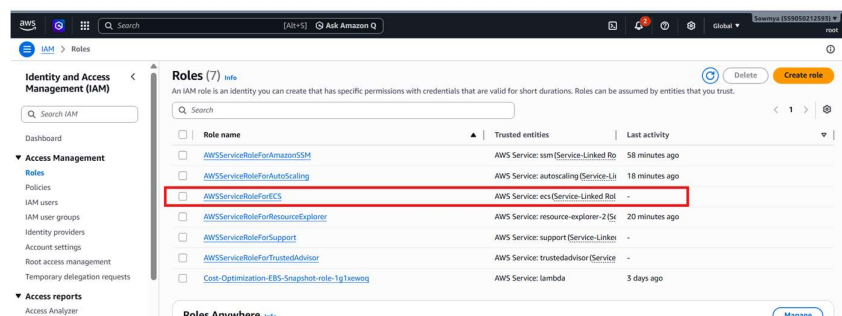
Architecture flow diagram



Step-1: Go to AWS console->ECS before that verify if you have valid IAM roles i.e AWSServiceRoleForECS. You can also add this via wsl with following commands

```
aws iam create-service-linked-role --aws-service-name ecs.amazonaws.com
```

Run only if AWSServiceRoleForECS does not already exist.



Step-2: Create a cluster and task definition along with Task Execution Role: ecsTaskExecutionRole

This role allows ECS tasks to pull images from ECR and send logs to CloudWatch.

Also, Assign Public IP = Enabled

The first screenshot shows the 'Clusters' page in the Amazon Elastic Container Service console. A green banner at the top indicates that the cluster 'soemmy-demo-cluster1' has been created successfully. The 'Clusters (1)' section shows a table with one cluster: 'soemmy-demo-cluster1' with 0 services, 0 tasks running, 0 EC2 instances, and no default capacity provider strategy.

The second screenshot shows the 'Create new task definition' page. The 'Task definition configuration' section has 'Task definition family name' set to 'soemmy-demo-task'. Under 'Infrastructure requirements', 'Launch type' is set to 'AWS Fargate'. 'OS, Architecture, Network mode' is set to 'Linux/x86_64' and 'Network mode' is 'awsvpc'. 'Task size' is set to 'CPU' and 'Memory' is '1 GB'.

The third screenshot shows the 'Container - 1' details section. 'Container details' shows 'Name' as 'soemmy-container' and 'Essential container' as 'Yes'. 'Image URI' is '559050121093.dkr.ec2.ap-south-1.amazonaws.com/soemmy-demo-nginx:latest'. 'Port mappings' shows a mapping for port 80 using TCP protocol and 'container-port-protocol' as the port name, with HTTP as the app protocol.

Step-3: Go to the cluster you have created and you will services. Click on create.

Task definition successfully created
sowmya-demo-task-1 has been successfully created. You can use this task definition to deploy a service or run a task. [View task definition](#)

Create service

Service details

Task definition family
Select a task definition family. To create a new task definition, go to [Task definitions](#).

sowmya-demo-task

Task definition revision [Learn](#)
Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.

Q, 1

Service name
Assign a service name that is unique for this cluster.
Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster.

sowmya-demo-service

Environment [AWS Fargate](#)

Existing cluster
sowmya-demo-cluster-1

Compute configuration - advanced

Compute options [info](#)
To manage task distribution across your compute types, use appropriate compute options.

☒ **Capacity provider strategy**
Specify a launch strategy to distribute your tasks across one or more capacity providers.

☐ **Launch type**
Launch tasks directly without the use of a capacity provider strategy.

Capacity provider strategy [info](#)
Select when your cluster default capacity provider strategy or select the custom option to configure a different strategy.

☒ Use cluster default
No default capacity provider strategy configured for this cluster.

☒ **Use custom (advanced)**

Capacity provider [Base](#) [info](#) [Weight](#) [info](#)

FARGATE 0 1

[Add capacity provider](#)

You can add up to 1 more capacity provider strategy item.

Platform version [info](#)
Specify the platform version on which to run your service.

LATEST

[Troubleshooting configuration - recommended](#)

Deployment configuration

Scheduling strategy [info](#)

☒ **Replica**
Place and maintain a desired number of tasks across your cluster.

☐ **Daemon**
Place and maintain one copy of your task on each container instance.

Desired tasks
Specify the number of tasks to launch.

1

Availability Zone rebalancing [info](#)
☒ **Turn on Availability Zone rebalancing**
Amazon ECS automatically detects Availability Zone imbalances in task distributions across an ECS service, and evenly redistributes ECS service tasks across Availability Zones.

Health check grace period [info](#)
0 seconds

Deployment options

Deployment failure detection [info](#)

Networking

VPC [info](#)
Select a VPC to use for your Amazon ECS resources.

vpc-084b03ae7150b56c

[Create a new VPC](#)

Subnets
Choose the subnets within the VPC that the task scheduler should consider for placement.

☐ Choose subnets

☒ **Clear current selection**

subnet-0a081f9c5c55f27 X subnet-0a084c150b0ff735 X subnet-0a304c5600209d1f X

ap-south-1a 172.31.0.0/20 ap-south-1a 172.31.16.0/20 ap-south-1a 172.31.32.0/20

Security group [info](#)
Choose an existing security group or create a new security group.

☐ Use an existing security group

☒ **Create a new security group**

Security group details
Specify the configuration to use when creating the new security group.

Security group name
ecs-03b9d9d8

Security group description
Created in ECS Console

Inbound rules for security group
Add one or more ingress rules for your security group.

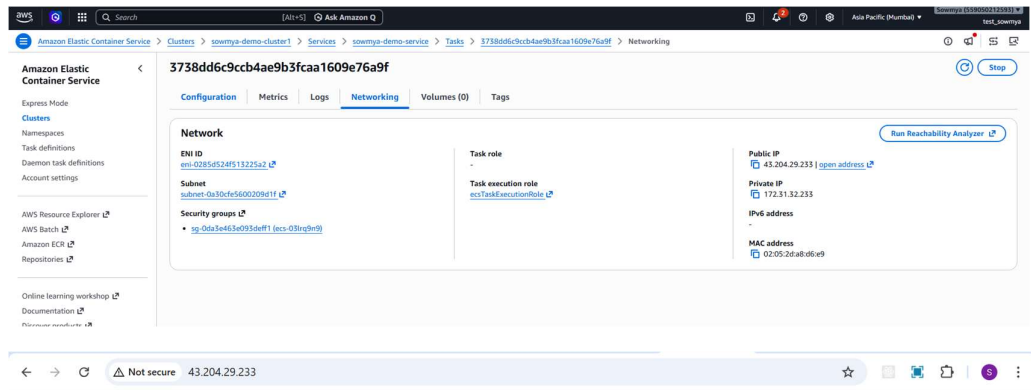
Type [Protocol](#) [Port range](#) [Source](#) [Values](#)

HTTP TCP 80 Anywhere 0.0.0.0/0

[Add rule](#)

Error: A valid port or port range between 0 and 65535. For example: 80 or 0-1023.

Step-4: Once service is created successfully go to service->select service created->Tasks->networking and copy the public IP and paste in web browser you will be able to access the page successfully.



Hello from ECR and ECS!

Deployed by Sowmya

Now, do not forget to delete the services once completed.

Cleanup

1. Delete ECS Service
2. Delete ECS Cluster
3. Delete ECR Repository
4. Deregister Task Definitions
5. Delete CloudWatch Log Groups (optional)